

Full Length Article

Formation of subsurface Cu-O-Si system through laser-induced plasma-assisted copper penetration for fabricating robust adhesive copper wire on glass substrate

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ABSTRACT

This study reports an approach, laser-induced plasma-assisted penetration (LIPAP), to facilitate copper diffusion to the glass interior using a femtosecond laser. TEM images and EDS analysis showed copper particles could diffuse into the glass up to 300 nm from the surface. The diffusion was driven by the local high temperatures generated by consecutively incident highly repetitive laser pulses. Furthermore, the HR-TEM images showed that, between the surface copper layer and the internal amorphous SiO₂, there were crystalline regions of copper nanoparticles and copper oxides and an amorphous zone, belonging to the Cu-O-Si system, through which, the surface copper layer was firmly connected with the glass substrate. The LIPAPed copper layer then served as a patterning seed for electroless-plating of highly conductive copper wires. Consequently, copper wires with a resistivity of $9.9 \times 10^{-8} \Omega\text{m}$ were obtained. Finally, a scratch adhesion test was performed to quantify the adhesion of the resulting copper films. Results showed that copper remained adhered to the surface before reaching the ultimate glass crush load of 30 N, corresponding to normal scratch stress of 240 MPa. Accordingly, based on the LIPAP-formed special subsurface nanostructures serving as the mechanical anchoring basis, highly conductive and adhesive electroless-plated copper wires were achieved.

1. Introduction

Glass has many favorable application properties such as low electrical loss, low dielectric constant, high electrical resistivity, high corrosion resistance, high stiffness, and adjustable coefficient of thermal expansion [1]. As a result, it has always been an unlimited and innovative material and an essential component of numerous products such as MEMS sensors [2], photonics [3], microfluidics [4], fan-out wafer-level packaging [5], Radio Frequency (RF) [6], and so on. As a substrate element, glass has unique advantages in flatness, transparency, heat resistance, tear resistance, and low deformation rate under high working temperatures. Thus, one of its promising applications is printed circuit board (PCB), the so-called Glass PCB, and it has been effectively used in the LED, 5G, LCD, etc. In circuit board applications, patterned metal wires, mainly copper wires, on glass are essential. The excellent physical and chemical stability of glass, however, makes the metallization of glass substrates very challenging, especially when high electrical conductivity and excellent metal/glass adhesion must be considered

simultaneously.

The common methods used in industry to metalize copper on glass are electrodeposition and electroless plating. Traditionally, these methods require many steps such as wetting, sensitization, photo-oxidation, activation, and metallization [7]. Although several methods have been proposed to speed up or simplify these processes, the complexity of these steps remains challenging [8–10]. Besides, as a circuit board application, copper/glass adhesion is essential. Failure to remain intact and adherent of circuits to the substrate can result in serious operation issues with economic losses, such as interruptions in signal transmission and damages to circuit boards, devices, and even systems. Therefore, the development of appropriate joining methods between glass and metal has always been a research topic with application importance [11].

One promising alternative for glass metallization is the LIPAA, laser-induced plasma assisted ablation. This method is based on the laser-induced backward transfer (LIBT) [12], where the donor and acceptor are not in contact and the surface topography, such as flatness and

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roughness, does not require special attention. LIPAA was initially proposed for micromachining of transparent substrates such as quartz, glass, and sapphire [13–20]. By adjusting the process parameters of laser and arrangement, such as pulse energy, pulse duration, pulse repetition rate, beam scanning speed, and the gap between the donor (metal) and the acceptor (substrate), metal particles could also be successfully deposited on the substrate without machining the substrate. Hanada et al. [21] deposited Cu wires on glass by LIPAA using various types of pulsed lasers with pulse durations of nanoseconds, picoseconds, and femtoseconds, respectively. The copper/glass bonding strengths for the resulting LIPAAed copper films were performed by the tape adhesion test. Results showed that the film formed by the nanosecond laser exhibited the strongest adhesion. Under this framework, Zhang et al. [22] employed a nanosecond laser to deposit a copper seed layer on a glass substrate for electroplating routing. The resulting adhesion of the electroplated film was examined by two methods: circulation ultrasonic cleaning in ethanol solution and strip pulling. They found some deep cracks on the glass substrate around the copper layer and speculated this was due to the interaction of the long laser pulses with the substrate during the LIPAA process. Lu et al. [23] used LIPAA to deposit copper wires on a sapphire substrate. They discussed the influence of different processing parameters on the final electrical resistivity performance of the copper wires. Xu et al. [24] fabricated graphite patterns on a glass substrate by LIPAA and further electroless plating nickel for electrofluidic devices.

The above literature review shows that LIPAA could form copper films with sufficient adhesion on glass substrates to withstand the tape strip pulling and ultrasonic agitation tests. However, quantitative descriptions of copper/glass adhesive strength remain unclear. Moreover, the interfacial nanostructure and corresponding adhesion mechanism have not been qualitatively revealed, either. Besides, within the range of existing research process parameters, the femtosecond laser LIPAAed copper layer could not even stand the tape strip pulling test. The potential reasons deserve further exploration. Within the framework of LIPAA, this study employs a femtosecond laser to deposit a copper layer on the glass surface. By well-adjusting process parameters, the layer is then heated by the energy of consecutive high-repetition femtosecond laser pulses, which facilitate the copper to penetrate the glass, creating special subsurface nanostructures that appear to favor a firm bond with the copper layer on the glass surface. Particular emphasis is placed on diffusion assistance, this approach is called laser-induced plasma-assisted penetration, LIPAP. The mechanical and electrical properties of the LIPAPed and the subsequent electroless-plated copper layers are studied. The nanostructures of the copper/glass interface are qualitatively examined based on Scanning Electron Microscope (SEM) and Transmission Electron Microscope (TEM) images and semiquantitatively analyzed using elemental compositions from Energy Dispersive Microanalysis (EDS). Finally, a scratch adhesion test is employed to quantify the bonding strength of the resulting electroless-plated copper film.

2. Experimental details

2.1. Laser processing system

The employed femtosecond laser is based on a diode-pumped fiber amplifier system (Amplitude, Satsuma HP²) which operates at the central wavelength of 1030 nm and the pulse duration of this laser is adjustable from 300 fs to 10 picoseconds. The repetition rate of the laser is tunable from 1 Hz to 500 kHz, and its maximum pulse energy output is 40 μ J. The irradiated laser beam, oscillating with a Gaussian-distributed electric field, was guided by a series of optics consisting of multiple mirrors, an aperture, and a beam expander, and was finally transmitted into a Galvanometer scanner (LASEA, LS-Scan XY10), as shown in the schematic diagram shown in Fig. 1. The focused laser beam from the scanner has a spot diameter of 34 μ m and a depth of focus (DOF) of 1 mm.

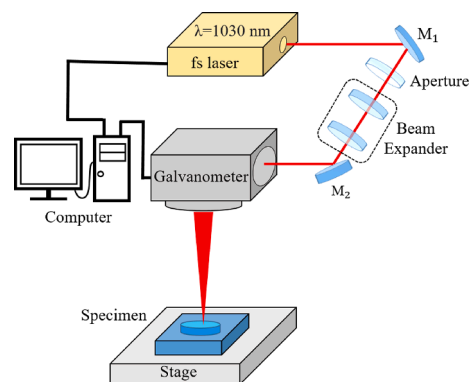


Fig. 1. Schematic of the laser processing system.

Referring to Fig. 2, for the LIPAP process, the acceptor substrate was a borosilicate glass (Corning, EAGLE XG), with a thickness of 200 μ m. The donor was a copper plate, and the copper purity was more than 99.9%. The spacing between the glass substrate and copper plate was fixed at 70 μ m by a spacer. Then, the LIPAP process was performed to fabricate copper wires on the glass substrate, which was accomplished by using a computer to simultaneously control the laser irradiation and beam scanning of the scanner. Before laser irradiation, both the glass substrate and the copper plate were first pre-cleaned with acetone and then rinsed in isopropanol with ultrasonic agitations for 30 mins.

2.2. Copper electroless plating

The electroless plating of copper was accomplished using commercially available solutions formulated by Teamly Chemicals, Taiwan. The main solution, named ECM-60, was prepared by mixing 600 ml deionized (DI) water with 1.75 ml ECM-60-1, 140 ml ECM-60-2, 4 ml ECM-60-3 and 8 ml ECM-60-4. Here, ECM-60-1 acted as a copper ion provider and contained 3.5 g of copper ions per liter, ECM-60-2 served as a chelate agent and contained mainly ethylenediaminetetraacetic acid (EDTA), and ECM-60-3 and ECM-60-4 both mainly contain stabilizers.

First, the ECM-60 was heated up to 60 $^{\circ}$ C, then 4.8 ml ECM-120N and 35.6 ml ECM-120H were added to the solution. The ECM-120N here mainly contained sodium hydroxide (NaOH) to make the solution alkaline and control the PH value of the solution at 12, while the ECM-120H mainly contained Formaldehyde (HCHO) as a reducing agent. After the LIPAP, the specimen was immersed in the abovementioned solution for 30 min, then washed with DI water.

2.3. Characterizations

We examined the geometry, surface morphology, and cross-sectional profile of the as-prepared, the LIPAPed, copper wires and the joined glass substrate by a laser scanning confocal microscope (Keyence, VK-X1000) and a scanning electron microscope, SEM (JOEL, JSM-7000F). The elemental compositions around the copper/glass surface were examined using X-ray photoelectron spectroscopy (XPS). A dual-beam focused ion beam, FIB (FEL, Versa 3D), was used to slice samples for the cross-sectional observation and analysis of the microstructures around the copper/glass junction area. The employed high-resolution transmittance electron microscope (HR-TEM) in this study was JEOL, JEM-2100.

The resulting electrical resistance of the electroless-plated copper wires was measured using a source measurement unit (Keithley, Keithley 2410) with the two-point measurement method, the resistivity was further calculated as the dimension of the copper wires were obtained. Finally, we quantified the adhesion strength of the copper wires on the glass substrate using a scratch adhesion tester (Anton Paar, RST³).

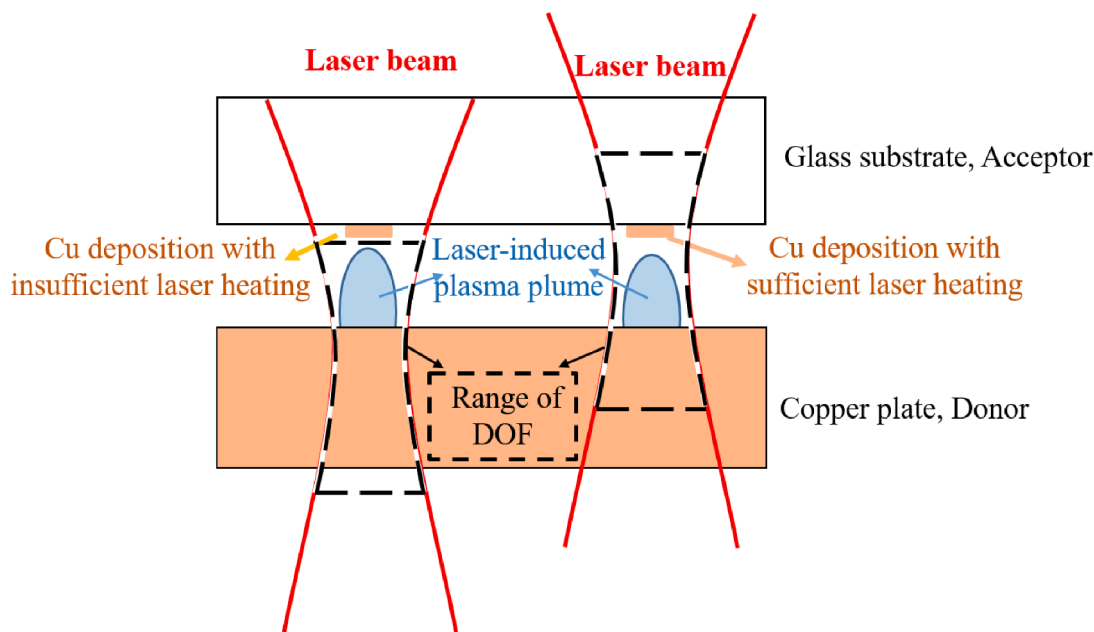


Fig. 2. Schematic showing cross-sectional view of two DOF arrangements.

3. Results and discussion

3.1. Mechanism of LIPAP process

The process of joining copper to glass through the approach of laser-induced plasma is first illustrated. Since glass is highly transparent, with a transmissivity larger than 90% to the light of the processing laser (operating at the wavelength of 1030 nm), the incident laser beam mainly passed through the glass substrate and was intercepted by the copper plate. Besides the portion of reflection, the absorbed laser energy, still had a high instantaneous peak power of about 27 MW, resulting in rapid copper ablation around the laser hit site. The ablated particles were with elevated temperature and part of them were ionized. With successive interactions with the subsequent incoming high-repetition laser pulses, the degree of ionization of the ablated particles was continuously increased. In addition, part of the surrounding ambient gases was ionized as well due to their frequent collisions with the ablative particulates. Consequently, a plasma plume was formed above the copper plate, as shown in Fig. 2. When the plume reached the acceptor, a portion of the substrate surface might be etched away due to the bombardment of the high-energy plasma. Depending on processing parameters, a thin copper film might also form at the etch site, and if the film was thick enough, it could act as an opaque mask, absorbing energy from the succeeding incident laser pulses and assisting the etching process. This is the so-called laser-induced plasma-assisted ablation, LIPAA, which is an alternative to the aforementioned microdrilling of transparent substrates.

However, the objective here is not to deposit copper on the glass as a mask for drilling, but to form a patterned seed copper layer with strong mechanical anchors on the glass surface for the subsequent electroless plating of copper electrodes or circuits. The mechanism of LIPAP is illustrated as the following. Under the current experimental setup, to successfully penetrate copper into the glass substrate through the approach of LIPAP, the DOF of the laser-focused beam had to cover both the bottom surface of the glass substrate, acceptor, and the top surface of the copper plate, donor. Fig. 2 schematically illustrates two settings with different DOF coverages. Our studies showed that if the region of DOF was off the glass bottom surface and only covered the copper plate, that is, the focal plane was closer to the copper top surface, as shown in the left schematic of Fig. 2, the deposited copper film could not adhere

strongly to the glass substrate and the resulting deposited film was easily removed within 30 min of ultrasonic agitation in water. However, using the same set of laser processing parameters, as the DOF covered both the glass substrate and the copper plate, as depicted in the right schematic of Fig. 2, an electroless-plated Cu film with very strong adhesion could be achieved.

The two arrangements differed in the power intensity at which the deposited copper film was hit by the subsequent incident laser pulses. For the left arrangement, the deposited copper film was unable to capture sufficient laser energy from the subsequent incident laser pulses for heating to facilitate copper diffusion. However, as the region of DOF covered both the surfaces of glass and copper, the arrangement on the right, more laser energy was intercepted by the film, which heated the glass and facilitated the diffusion of copper toward the glass interior. The subsurface and surface coppers were interconnected that acted as metal seeds for the later electroless plating. Due to the subsurface mechanical anchoring effect, the resulting copper film had strong adhesion to the substrate, which could not only withstand ultrasonic vibration in water but also survive after scratch adhesion test under high scratch load. A further quantitative discussion of adhesion will be presented later.

3.2. Morphology and resistivity of LIPAPed and electroless-plated copper wires

Fig. 3(a) shows the top view of an optical microscope (OM) image of a LIPAPed copper wire on the glass substrate. The laser operated with a pulse energy of 8 μJ , a pulse repetition rate of 100 kHz, and a beam translation speed of 4 mm/s. The pulse overlap, PO, was 99.88%. Here PO is defined as $1 - \frac{v}{fD}$, where v , f , and D represent beam translation speed, pulse repetition rate, and beam spot diameter, respectively. Unless otherwise stated, the LIPAP process used this set of laser process parameters, which were later shown to be an appropriate one. A high percentage of PO is critical for reaching and maintaining a high local temperature at the copper/glass boundary. Some areas on the surface of the copper wire appeared dark red and black, indicating the formation of CuO and Cu₂O on the surface. This result was predictable because it was obtained through high-temperature plasma processing in an atmospheric environment that accelerated the process of copper oxidation. Fig. 3(b) is an SEM image for a more detailed observation of the wire

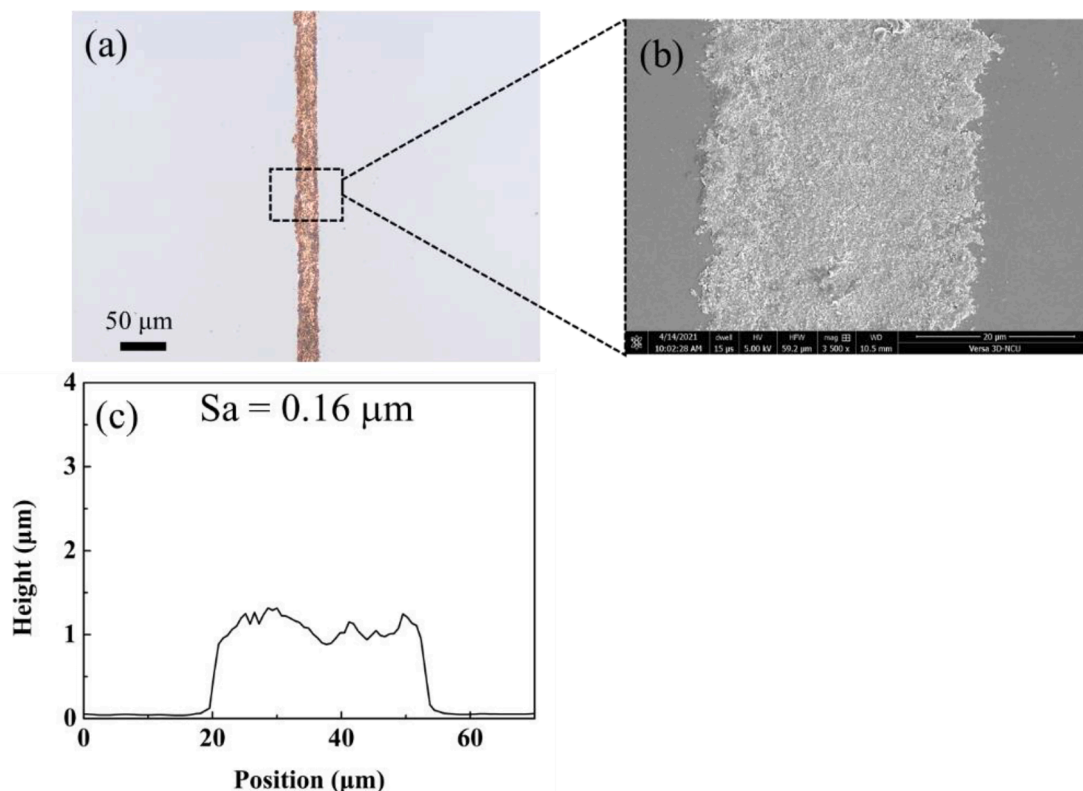


Fig. 3. Surface morphology of a LIPAPed copper wire. (a) OM image of the copper wire. (b) SEM image of the copper wire. (c) The cross-sectional surface profile of the copper wire.

surface morphology. It looks like the wire surface was completely covered with copper, but there were still some unevennesses. Fig. 3(c) shows a cross-sectional surface profile of the wire, indicating that it had a width of about 33 μm and a height of around 1.2 μm, while the measured surface roughness was $S_a = 0.16 \mu\text{m}$.

To investigate and qualify the chemical compound of the copper wire, we present the Cu 2p XPS spectrum in Fig. 4. Cu 2p energy level was composed of two characteristic peaks of two core level binding energies of Cu 2p_{1/2} and Cu 2p_{3/2}, at 952.6 eV and 932.8 eV, respectively. The shake-up satellite peaks at joining energy were about 8 eV and 10 eV higher than the 2p_{3/2} peak. The extracted peaks at 932.7 eV and 952.5 eV corresponded to the Cu 2p_{3/2} and Cu 2p_{1/2} peaks of Cu⁰ and Cu⁺ species, respectively. It is challenging to differentiate Cu⁰ and Cu⁺ from the XPS data because their peaks were highly overlapped, but the strong and broad signals of the peaks indicated the coexistence of the two species [25]. The extracted peaks at 934.2 eV and 954.3 eV corresponded to Cu 2p_{3/2} and Cu 2p_{1/2} of Cu²⁺ species, respectively. Also, the shake-up satellite peaks indicate the presence of Cu²⁺ species

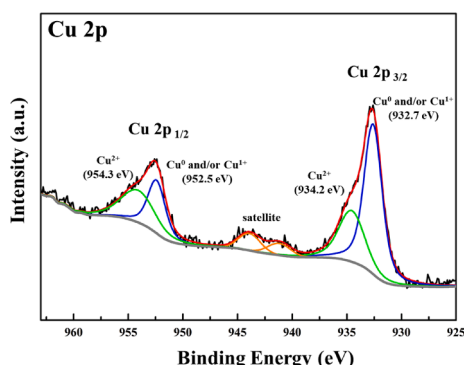


Fig. 4. XPS spectrum of the LIPAPed copper wire.

[26]. The information from the XPS spectra thus indicated the LIPAPed copper wire consisted of pure Cu, Cu₂O, and CuO.

For applications such as conductive circuits and electrodes, the fabricated copper wire must exhibit small resistance. The measured electrical resistivity from the LIPAPed copper wires was about $8 \times 10^{-4} \Omega\text{m}$, ten thousand times higher than that of the bulk copper. The resistance was too high for the application of glass substrate routing. The electrical performance could be improved if processed under vacuum or inert gas conditions to reduce the degree of oxidation of Cu. However, considering the large area and low-cost fabrication process, it is preferable that all processes are performed under atmospheric ambient conditions. Therefore, we applied the technique of electroless plating of copper to further enhance the conductivity. Here the LIPAPed copper wire served as a patterned, metal seed layer. Since the CuO and Cu₂O compounds on the LIPAPed wire could be deoxidized in the copper deposition bath solution, the newly grown copper particles were bonded to the LIPAPed wire during the plating process. Fig. 5(a) gives the OM image of the copper wire after the electroless plating. The surface of the Cu wire was bright brown, indicating the presence of copper on the surface. The surface roughness was $S_a = 0.2 \mu\text{m}$, which was a little bit rougher than that before electroless plating. The key factor for regulating surface roughness was the rate of depositing the copper particles. Smoother surface roughness can be accessed by adjusting processing parameters to moderate the depositing rate, such as lowering the baths' processing temperature, dropping the PH value at the bath, or reducing the dose of the reducing agent [27]. This study focuses mainly on the approach of forming a stronger bonding between glass and copper and its plausible mechanisms. Therefore, the parameters for electroless plating were set to exceed the plating activation threshold to ensure that the process was prone to occur, and parameter optimization was not performed.

Fig. 5(b) shows the profile of the copper wire after the electroless plating. Compared with the previous LIPAPed one, the height and the

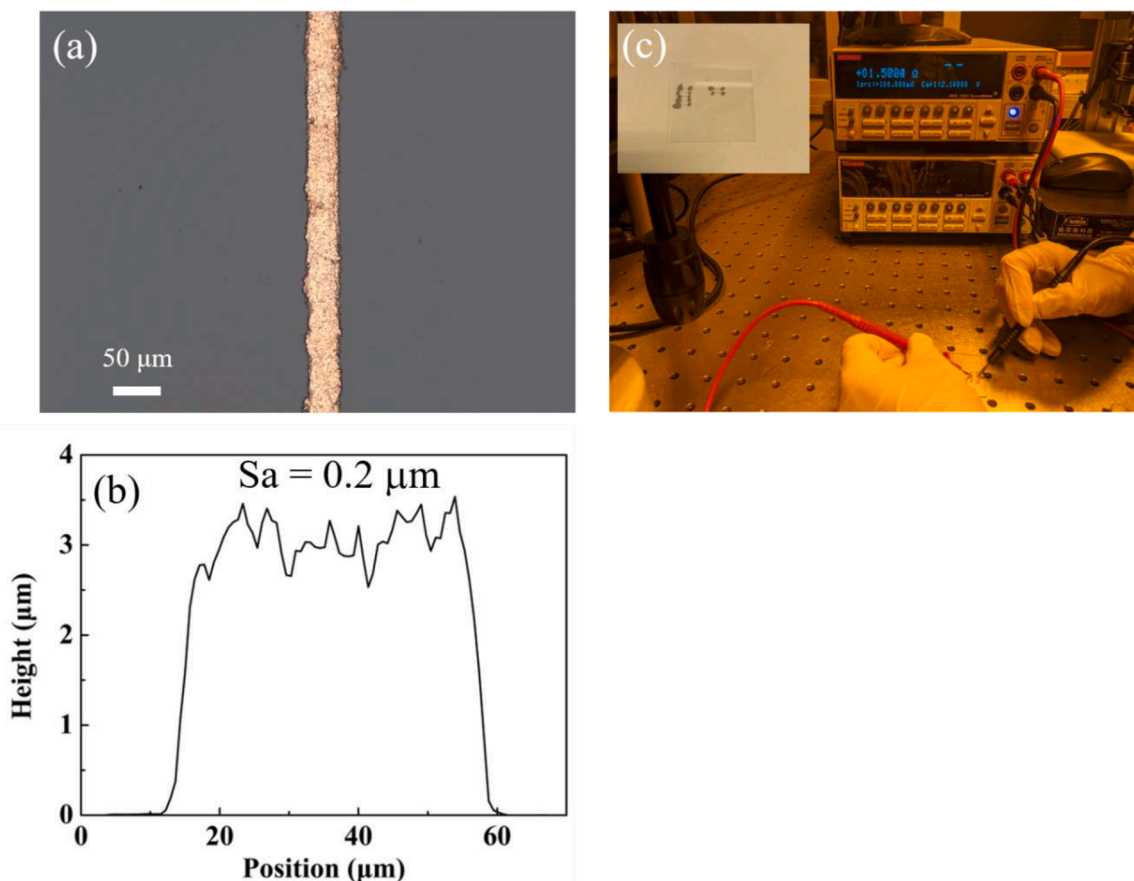


Fig. 5. (a) OM image of a copper wire after electroless plating. (b) Photo showing the measurement of electrical resistivity. (c) The cross-sectional surface profile of the electroless-deposited copper wire.

width were increased from 1.2 μm to 3 μm and from 33 μm to 42 μm , respectively. Fig. 5(c) shows the photo of the arrangement for electrical resistance measurement. To facilitate contact with the probes, the silver glue was applied to both ends of the copper wire. Then the resistivity was obtained according to $\rho = \frac{RA}{L}$, where ρ represents the electrical resistivity of the wire, R is the measured resistance, A represents the cross-sectional area of the wire, and L denotes the length of the wire. By averaging ten copper wires, the calculated resistivity was $9.9\text{A} \pm 0.33 \times 10^{-8} \Omega\text{m}$. This value was about six times that of bulk copper and was comparable to the resistivity obtained through general cooper electroplating. We demonstrated that, after electroless plating, the copper wire's electrical conductivity was greatly improved and could meet the requirements of electrode and circuit applications.

3.3. TEM observation and characterizations

To study the bonding structure of the copper/glass interface, understanding the characteristics of the structure and chemical composition at the nanoscale is necessary. Fig. 6 (a) shows the bright-field TEM image of the cross-sectional view around the copper/glass interface for the LIPAPed workpiece. Since copper nanoparticles are atomically heavier than glass, and possessed crystalline structures, causing more intense scatter and diffraction of the emitted electrons, demonstrating dark grey or black in the TEM image. From the image, in addition to those deposited on the glass surface, the copper nanoparticles that diffused into the glass interior via the LIPAP process were also detected, and because of that, the exact copper/glass interface became blurred. From the surface deposited copper layer to the glass interior, we selected four regions (Region A to Region D) in the TEM image for the Selected Area Electron Diffraction (SAED) pattern observation to examine the

interior copper particles and verify that they were formed by the LIPAP process. The images were taken using an aperture of 150 nm.

Fig. 6 (b) shows the spotted SAED pattern for Region A, centered on the copper deposited layer, indicating the presence of large copper particles with a single crystalline structure. There were two distinct spot patterns, identified as the Cu (1 1 1) and Cu (2 0 0) plane orientations and with the calculated interplanar spacings of 0.20 and 0.18 nm, respectively. The values agreed well with the face-centered cubic structure of Cu (JCPDS-ICDD no.71-4610, $d_{111} = 0.20883 \text{ nm}$, $d_{200} = 0.18085 \text{ nm}$). Fig. 6(c) shows the SAED pattern for Region B, an area including both the copper deposited layer and the glass interior. Compared with Fig. 6 (b), the pattern scattered spots were smaller which might be contributed by copper nanoparticles with smaller sizes or other Cu-based compounds, such as CuO and Cu₂O nanoparticles. Besides spots, there was an unobvious diffused ring at the center, indicating the detection of amorphous SiO₂ structures present in the glass matrix. The next region to be observed was region C, which was slightly further from the interface compared to Region B, and the corresponding SAED pattern was shown in Fig. 6 (d). There was one distinct ring, which corresponded to the Cu (1 1 1) plane with polycrystalline structure, contributed by the small-sized Cu nanoparticles. The SAED pattern also showed a diffused and thick ring corresponding to amorphous SiO₂. The SAED pattern for Region D, located even further from the interface, was presented in Fig. 6 (e) and demonstrated a strong diffused ring. This pattern indicated that only SiO₂ was detectable and no presence of copper nanoparticles.

The results in Fig. 6 show the presence of copper nanoparticles and copper-based compounds inside the glass, and the penetration depth of copper was at least 300 nm from the interface. The high temperature generated during copper deposition was presumed to be the driving force for copper diffusion. To acquire more information about the

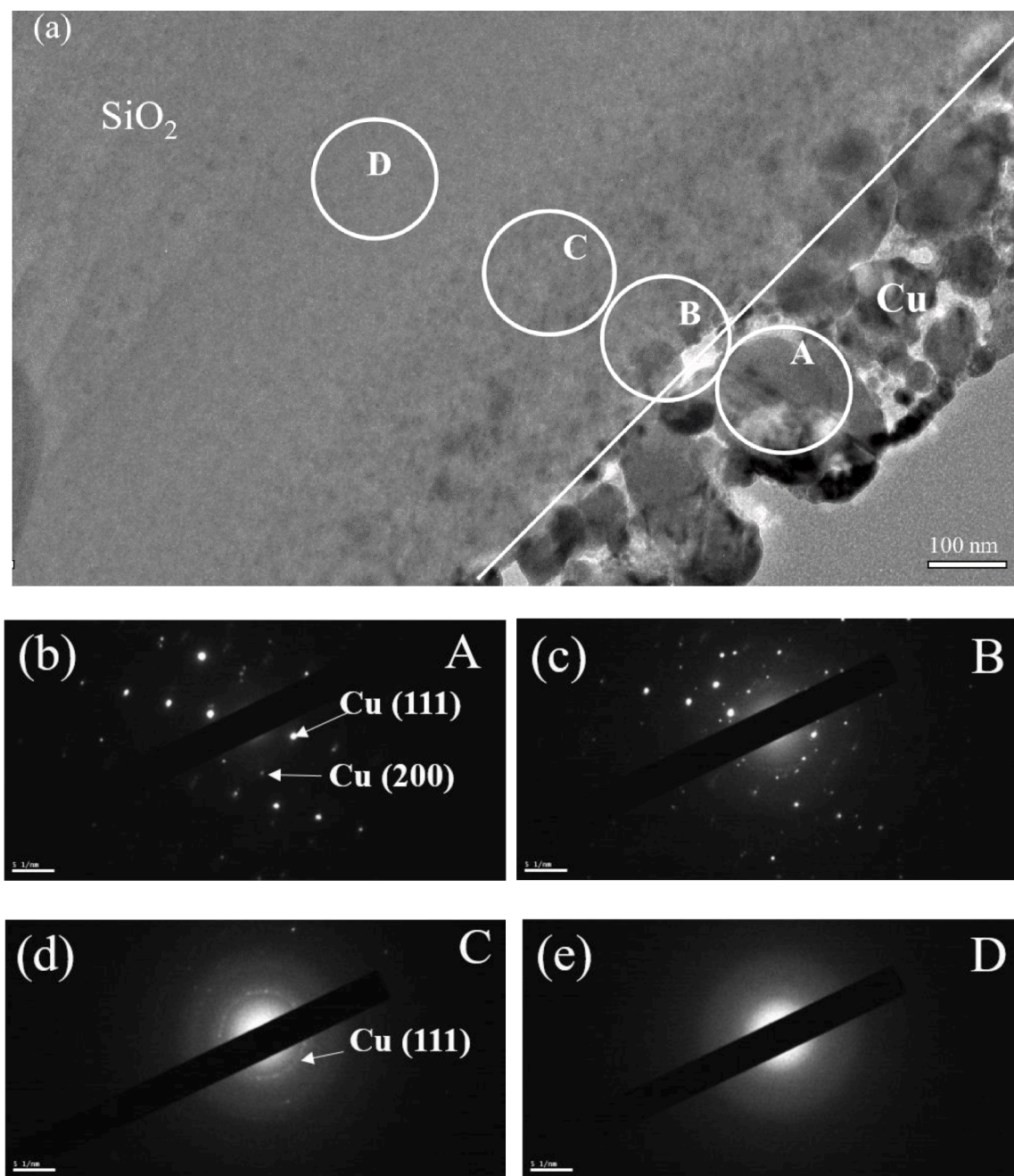


Fig. 6. The TEM and SAED images of the cross-sectional nanostructures at the copper/glass interface. (a) TEM image. (b) SAED pattern for Region A. (c) SAED pattern for Region B. (d) SAED pattern for Region C. (e) SAED pattern for Region D.

copper diffusion phenomenon, we applied energy dispersive spectroscopy (EDS) for qualitative and semi-quantitative analysis to identify the copper and silicon elements in the specimen.

Fig. 7 (a) shows an EDS mapping image of copper elements around the copper/glass interface of a LIPAPed workpiece. From the intensity of dot (green) distribution, it could be seen that the concentration of copper decreased gradually from the surface to the interior of the glass. It is also noted that the copper signal was ambiguous (dark) in some regions on the LIPAPed layer. These “voids” were due to the deposition of copper nanoclusters of non-uniform size during the LIPAP process. Fig. 7 (b) shows the path of the EDS line scan analysis, and the corresponding elemental concentrations of copper (black) and silicon (blue) elements along the scan path are shown in Fig. 7 (c). The orientation of the scanning was perpendicular to the interface, as indicated by the arrow symbol. Again, because of the voids, there were “valleys,” a lack

of copper, along the line even within the extent of the copper layer. As the path crossed the interface and penetrated the interior of the glass, the green line exhibited an exponential decay. Instead, the blue line, grew rapidly and then stabilized. There was a region where Cu and Si signals were both noteworthy, indicating the coexistence of the two elements, roughly in the red dashed box. Again, the range of this region was about 300 nm. We further investigated the nanostructures around the copper/glass interface by high-magnification HR-TEM images.

In Fig. 8, there were two distinct regions, a darker zone on the left and a lighter area on the right, composed of the single-crystalline Cu nanoparticles and amorphous SiO_2 , respectively. The lattice fringe of the nanoparticle crystalline was with a d-spacing of 0.2 nm that corresponded to the Cu (111) plane while, around the edge of the Cu nanoparticles, the lattice fringe showed a d-spacing of 0.245 nm that was from the Cu_2O (111) plane. The formation of the oxidated copper

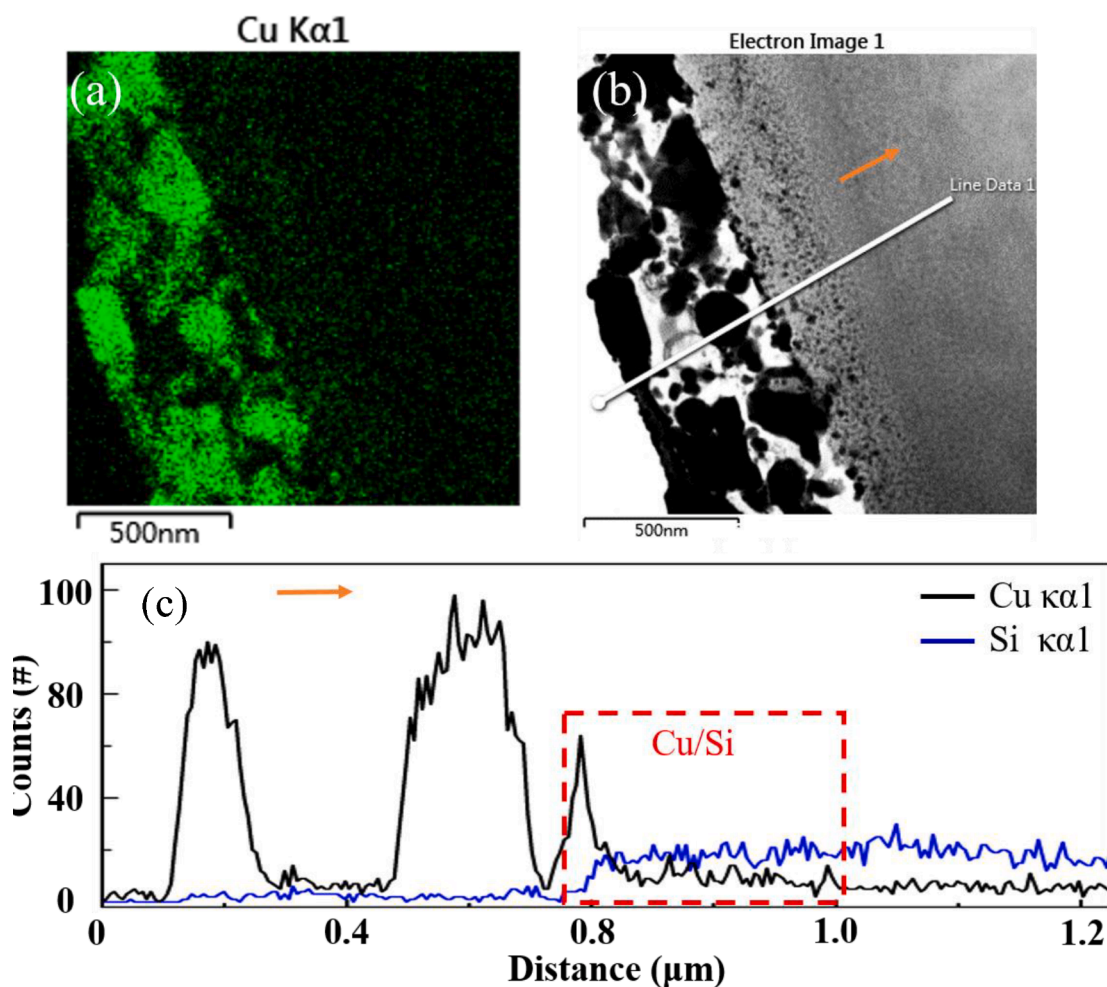


Fig. 7. EDS analysis of the copper/glass interface. (a) EDS mapping of the copper element. (b) TEM image of the EDS line scan region. (c) The intensities of Cu and Si elements along the scanned path. The red dash box represents the approximate depth of copper diffusion. Arrows in (b) and (c) indicate the scan direction. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

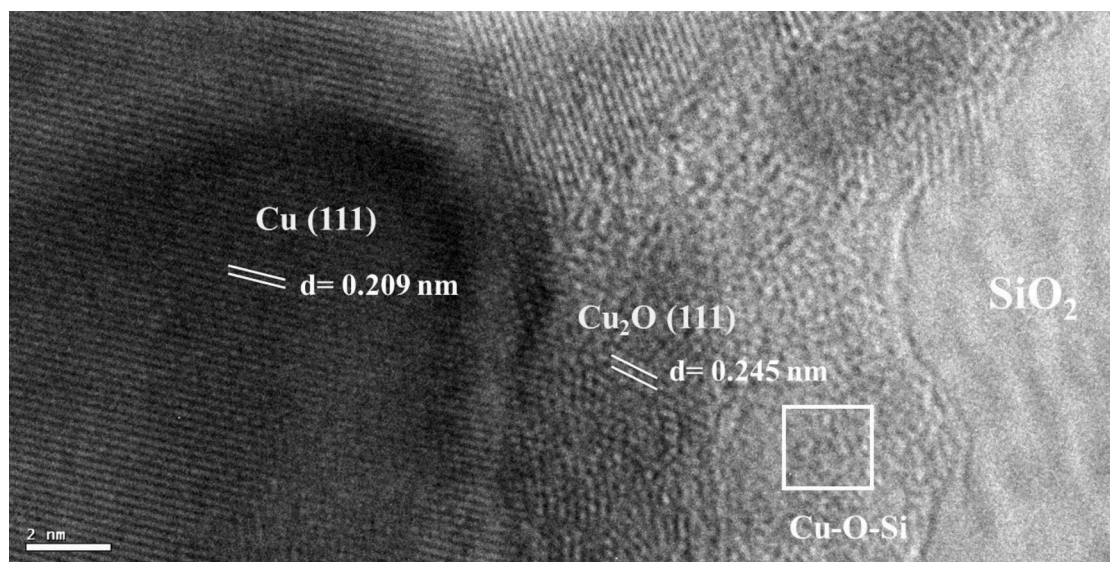


Fig. 8. HR-TEM image taken at the copper/glass interface. Single crystalline Cu and Cu_2O nanostructures were identified by the d-spacing calculation. An amorphous Cu-O-Si nanostructure was observed between the SiO_2 and Cu_2O , bonding Cu nanoparticles with the glass substrate.

nanoparticles was speculated due to the interactions of the entering copper particles, driven by diffusion, with the oxygen of the SiO_2 of the glass matrix. It was noted that between the Cu_2O crystalline and the amorphous SiO_2 , there was another amorphous region whose structure was different from that of SiO_2 . Based on the study of the phase equilibria of Cu-O-Si systems in equilibrium with the air and metallic copper by Hindayat et al. [28], this region was plausibly contributed by the Cu-O-Si system. With the presence of this system, the temperature around the interfacial region was expected to be higher than 1337 K during the LIPAP process because it is the system's eutectic temperature in equilibrium with the air [29]. The Cu-O-Si system, ubiquitously found between Cu nanoparticles and amorphous SiO_2 , provided the basis for a strong connection between the two substances. Consequently, the excellent adhesion strength between the glass substrate and the LIPAPed Cu layer could be achieved.

3.4. Scratch adhesion test of copper/glass bond strength

Adhesion testing is a valuable indicator for determining how well the deposited copper film is bonded to the glass substrate. There are several different tests to measure the bonding strength of the deposited from the substrate. Of these, the scratch adhesion test is popular because it provides a quantitative analysis of the deposited at different stages of failure. It is a destructive measurement that uses a diamond stylus to scratch across the coating surface down to the substrate surface, increasing load at a constant displacement rate until the coating fails. There are two characteristic failure values for a coating film on the substrate, Lc_1 , and

Lc_2 . Lc_1 is defined as the critical load of the film cracking that occurs in the early stage. Following the critical load, the Lc_2 is defined as the onset of the coating detachment, which quantifies the adhesion strength of the copper/glass interface.

The occurrence of Lc_2 can be determined from OM images, by observing the appearance of surface defects; or, be resolved by probing the changes in acoustic emission (AE) signal that varies detectably when the coating is detached. Both methods were presented in this study. On the other hand, Lc_1 is sometimes hard to visualize or identify through the signal. In this study, we employed a linearly increasing loading force ranging from 0.5 to 30 N at a rate of 24 N/min. The length of the scratch track was 4 mm. Since the radius of the employed indenter tip on the stylus was 200 μm , this was wider than the width of the manufactured copper wire. A Cu film with a width of 1 mm was fabricated for this measurement and it was achieved by fabricating adjacent but non-overlapping Cu wires. The scrapes were performed in the central portion of the film to avoid any possible edge effects.

The scratch adhesion test was performed on the electroless-plated film, with a thickness of about 3 μm . In the top portion of Fig. 9 was the OM image of the sample after the test, where the damaged, scratched region was indicated. In the lower part of the figure, both the corresponding AE signal and the penetration depth of the probe head as a function of the applied normal scratch load were displayed. The Lc_1 was assigned as 10.1 N from the AE signal. It was surprising that the surface of substrate glass did not even disclose until the end of the range of the test load, indicating Lc_2 was higher than the maximum preset load, which was 30 N in this test. Since the radius of the indenter tip was 200

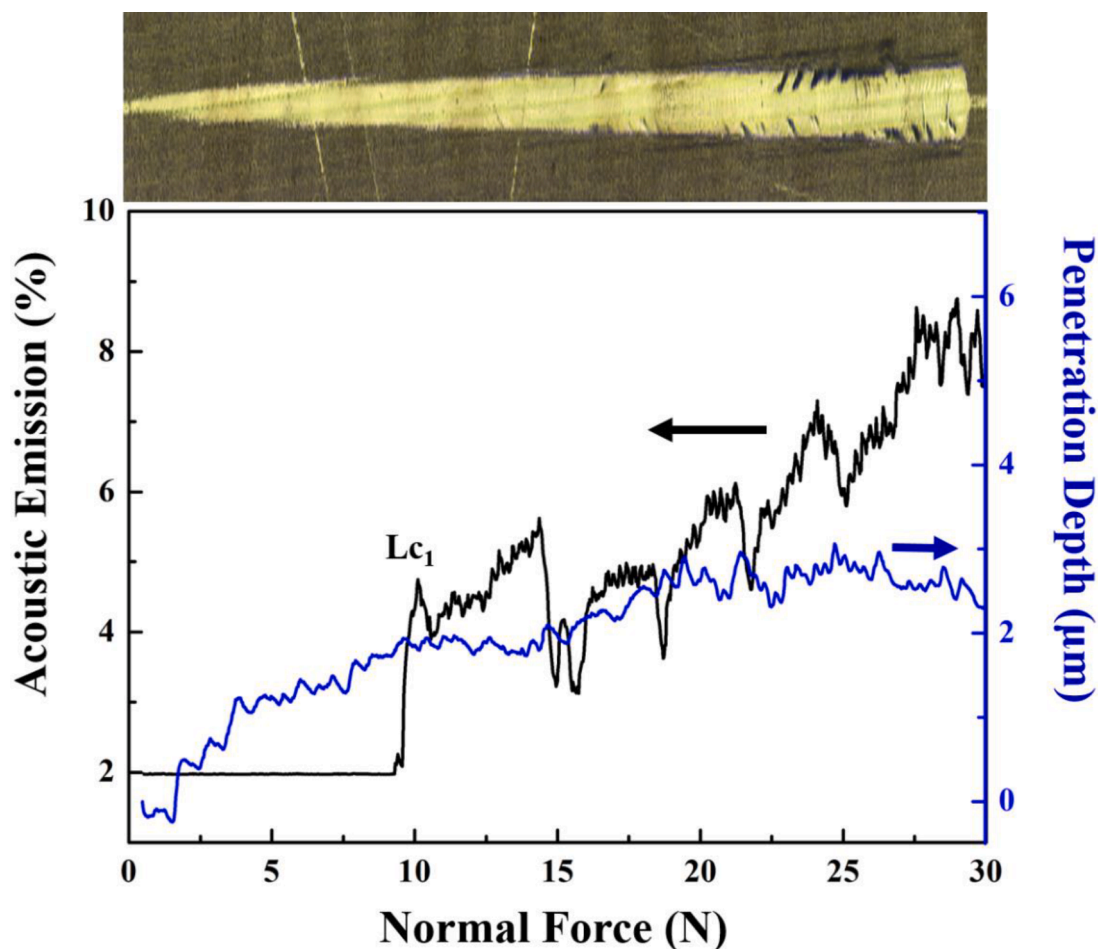


Fig. 9. Scratch adhesion test results. The upper portion was the top-view OM image of the sample after the test. The lower part showed both the acoustic emission (left vertical coordinate) and the penetration depth (right vertical coordinate) as a function of normal scratch load. The critical load of Lc_1 was shown to be 10.1 N. The Lc_2 was, however, unidentifiable till the load limit.

μm , the corresponding normal scratching stress of the force of 30 N was about 240 MPa. Any load above this value would crush the substrate glass, and once the substrate was broken, the indenter tip would hit the tester's stage and cause the tip to break. Thus, it demonstrated that the electroless-plated copper film, grown on the top of the LIPAPed copper seed layer, exhibited very strong adhesion strength to withstand scratch stress exceeding 240 MPa.

After passing the LC_1 , the AE signal increased roughly with the increase of the scratch force. However, when the force was larger than 15 N, it fluctuated substantially. This might be attributed to the presence of voids within the LIPAPed copper layer, as described previously. The corresponding OM image shows the scratched region increased with increasing force. For the lower scratching force range, the removed portion was mainly on the upper surface of the resulting film, which was primarily developed by electroless plating, and this process could form a relatively uniform film structure compared with that deposited by the LIPAP procedure. For the larger force range, the scratched region reached the lower portion of the film, mainly formed by the LIPAP process, where there was a less uniform and void structure, that made the AE signal complex and irregular.

The result of penetration depth as a function of scratch force indicated that the penetrated depth increased with the increasing force. When the scratch force was above 20 N, the penetration depth fluctuated between 2.5 and 3 μm , and wouldn't increase with the force increasing. Referring to Fig. 5(b), the thickness of the resulting film was about 3 μm . Because of the existence of the subsurface Cu-O-Si system introduced by the LIPAP process, it not only provided a solid base for the strong adhesion of the surface copper layer but also enhanced the strength of the glass surface to resist scratching force. As a result, no surface of the substrate glass was scraped off.

4. Conclusion

Within the framework of laser-induced backward transfer (LIBT) and through a suitable arrangement of laser processing parameters, the gap between the donor (copper plate) and acceptor (the glass substrate), and the location of the depth of focus of the field lens, this study reports a method that can simultaneously facilitate the diffusion of surface deposited copper to the interior of the glass substrate using a femtosecond laser, specifically laser-induced plasma-assisted penetration (LIPAP).

TEM images combined with SAED inspection and EDS elemental analysis showed that the LIPAP copper particles diffused to the interior of glass at a distance of at least 300 nm. HR-TEM images demonstrated the nanostructures of the subsurface copper-diffusion region, from the surface toward the glass interior, consisting of crystalline copper particles, polycrystalline copper oxides, and amorphous structures, which may belong to the Cu-O-Si system. This structure provided a bridge that firmly bonded the surface copper layer to the glass substrate.

The electrical resistivity of the LIPAPed copper layer was about $8 \times 10^{-4} \Omega\text{m}$, ten thousand times higher than that of the bulk copper. To further improve conductivity for circuit or electrode applications, the LIPAPed copper wires then served as the patterned metal seed for the subsequent process of electroless plating of copper. As a result, the resistivity was dramatically reduced to about $9.9 \times 10^{-8} \Omega\text{m}$, which was six times that of bulk copper and that could meet the practical application requirements.

The adhesion strength of the resulting copper layer on the glass substrate was quantitatively measured by the method of scratch adhesion test. Results showed there was still a copper layer that remained adhered to the substrate before the glass crush limit load of 30 N, indicating its interfacial adhesion could withstand scratching stresses above 240 MPa. Accordingly, a high-conductive copper wire with robust adhesion to the glass substrate was accomplished by the hybrid of LIPAP for patterning the seed metal layer and the electroless plating for

growing the resulting copper wires.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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